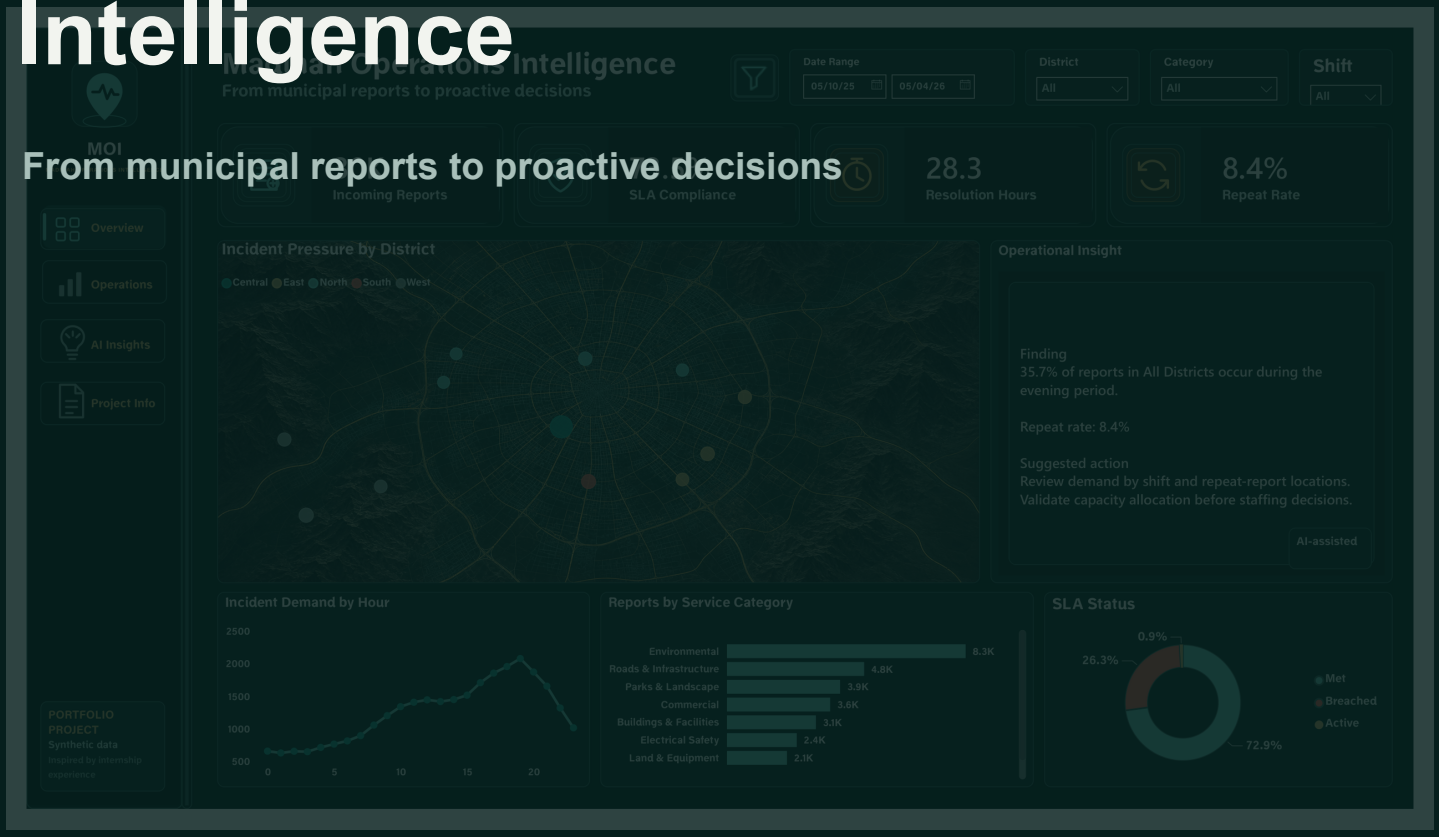
Madinah Operations Intelligence

From municipal reports to proactive decisions



BUSINESS INTELLIGENCE / OPERATIONS ANALYTICS / PREDICTIVE AI

An end-to-end portfolio solution inspired by internship experience at the Municipal Operations Monitoring Center in Madinah.

Yomna Alhejaili

Data Analytics, Business Intelligence & AI-Assisted Operations

PORTFOLIO CASE STUDY / 2026

01 Context & Business Challenge

The opportunity

Municipal operations generate a continuous stream of service reports. The challenge is not only to count them, but to translate demand, capacity and service-level performance into timely operational decisions.

PROJECT FACTS

Internship period
05 Oct 2025 - 05 Apr 2026

Dataset
30,000 synthetic municipal reports

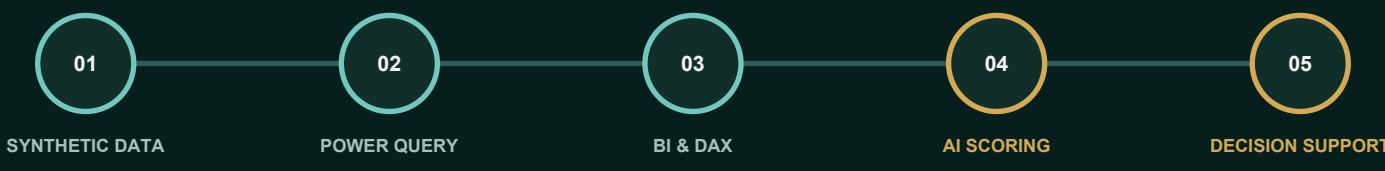
Domain
Municipal Operations Intelligence

Core operational questions

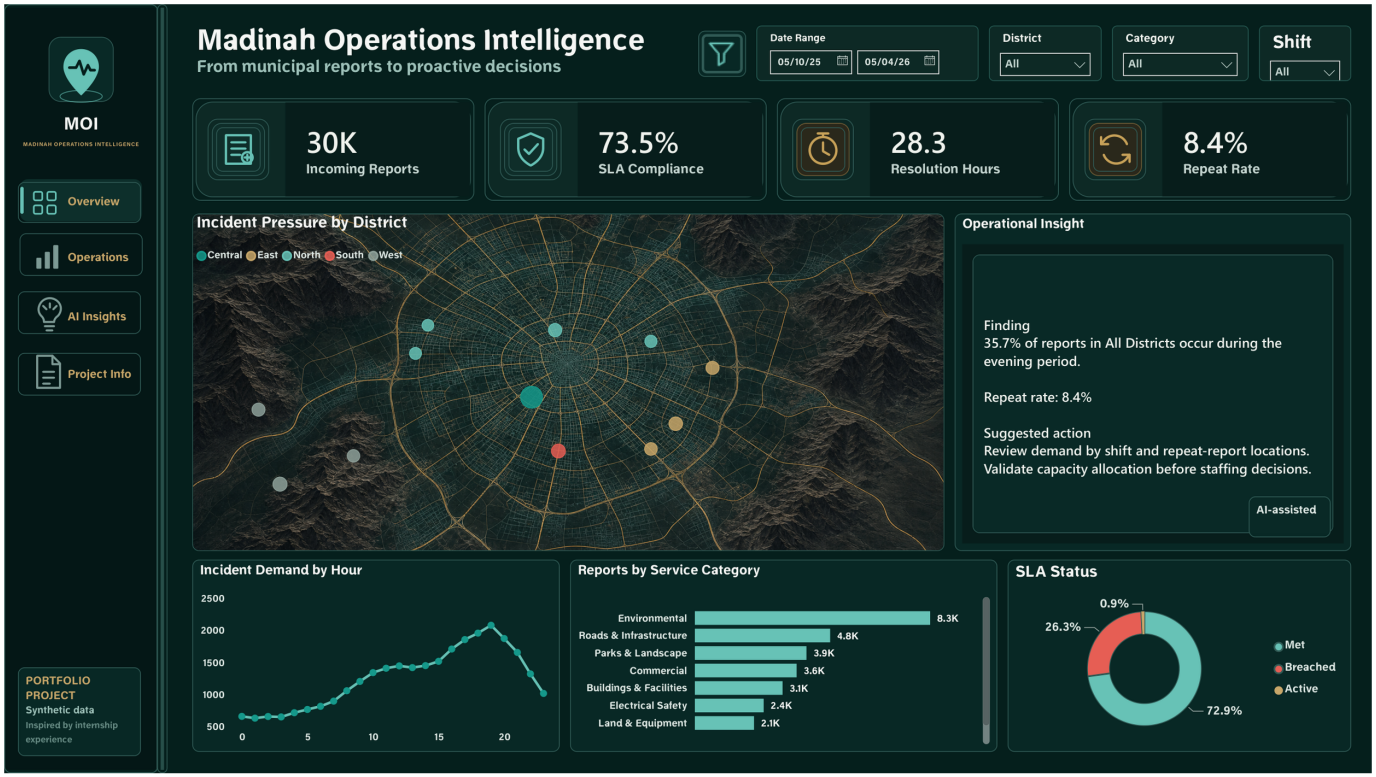
- Where is report pressure concentrated?
- How does demand compare with nominal capacity?
- Which districts and shifts are at risk of SLA breach?
- Which active reports should teams prioritize first?

DESIGN PRINCIPLE

Move from descriptive reporting to proactive, human-reviewed decision support.



02 Executive Overview



30K
INCOMING REPORTS

73.5%
SLA COMPLIANCE

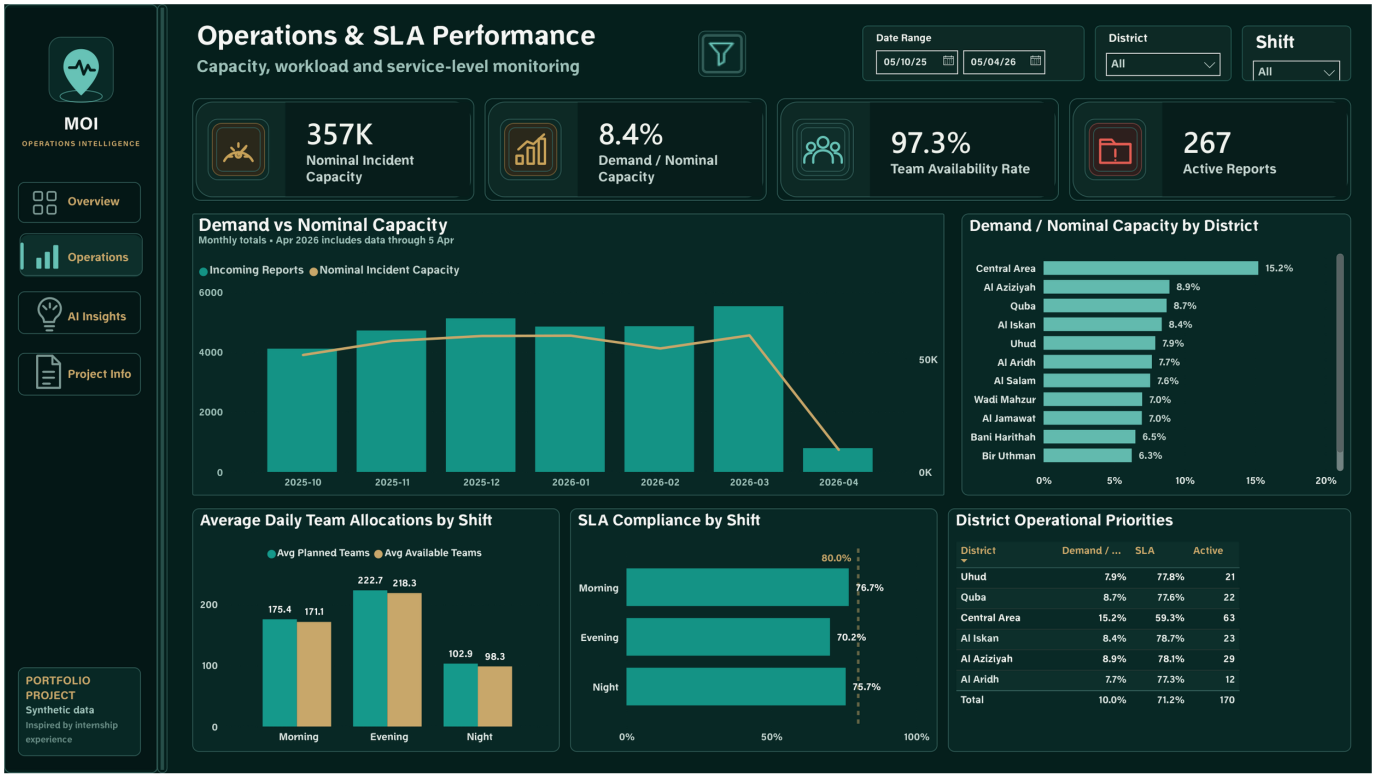
28.3h
MEDIAN RESOLUTION

8.4%
REPEAT RATE

What this page enables

- A single executive view of demand, SLA performance and repeat-report pressure.
- A spatial view that highlights report concentration around the Central Area.
- Hourly and service-category patterns that support resource planning.

03 Operations & SLA Performance



Operational interpretation

- Monthly demand is compared with nominal incident capacity.
- District demand / nominal capacity is compared; allocation is not operationally validated.
- Daily team allocations and SLA results support review before staffing decisions.

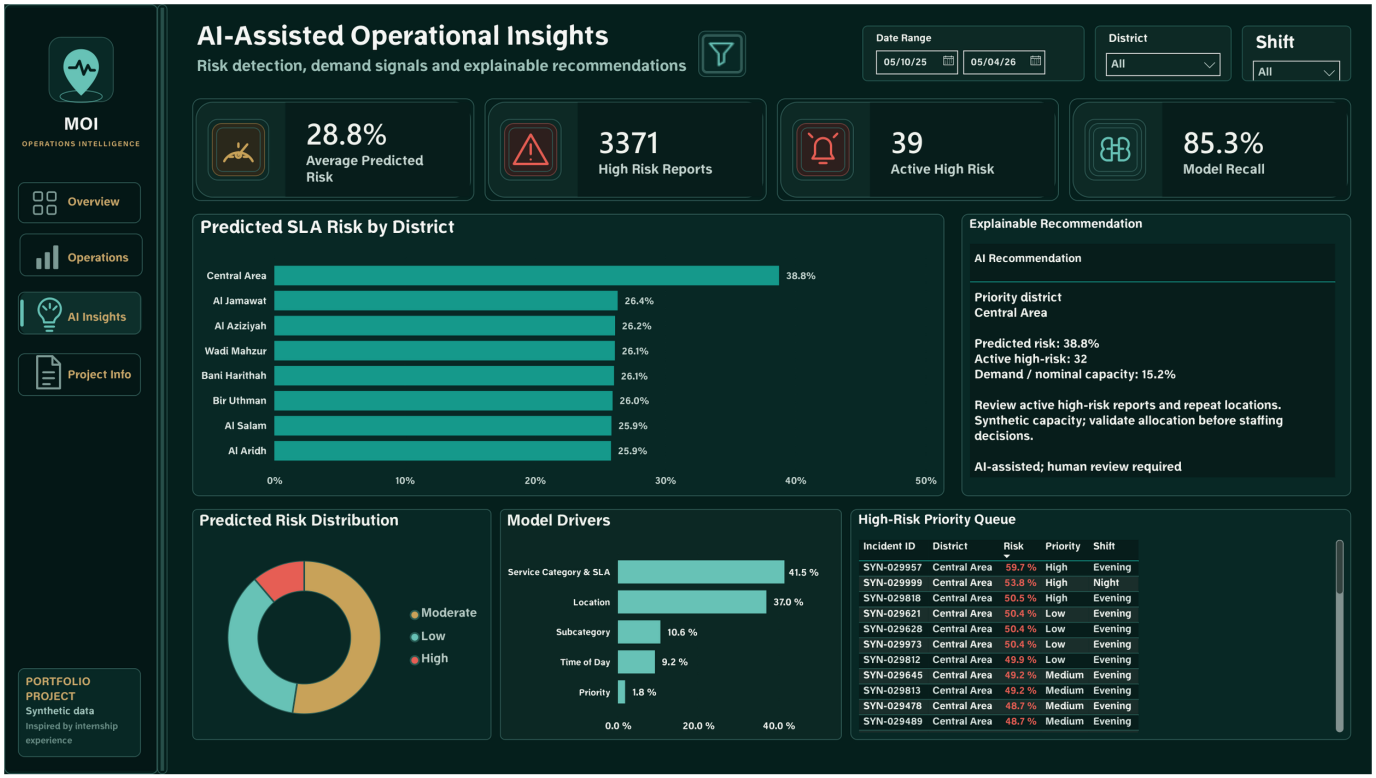
DECISION SIGNAL

Central Area nominal ratio: 15.23%, while current SLA compliance is 59.3%.

Recommended response

Validate capacity allocation and review active high-risk reports before staffing decisions.

04 AI-Assisted Risk Prioritization



68.7%
TEST ROC-AUC

85.3%
MODEL RECALL

24%
DECISION THRESHOLD

39
ACTIVE HIGH-RISK

MODEL APPROACH

Target: SLA breach before resolution
Models: scikit-learn Logistic Regression + AdaBoost
Pipeline: ColumnTransformer preprocessing
Validation: Labels known before each time cutoff

EXPLAINABILITY & GOVERNANCE

High risk >= 40%; review flag >= 24%.
All 30,000 rows scored, including training rows.
Staffing excluded; feature assumptions remain.
Retrospective scores; censoring limits apply.

AI-assisted - Human review required

05 Outcome, Governance & Next Steps

Business value

01

Visibility

A unified view of demand, location, service category and SLA performance.

02

Operational control

Demand / nominal capacity aids review; allocation remains unvalidated.

03

Proactive prioritization

Predicted risk helps teams focus on reports most likely to breach SLA.

Responsible use

- All source records are synthetic and privacy-safe.
- The model is a portfolio prototype and is not production-deployed.
- Retrospective scores include training rows; label-censoring limits remain.
- Predictions support - and do not replace - human operational decisions.
- A production path would add live integration, monitoring, drift checks and periodic retraining.

ABOUT THE AUTHOR

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